THESIS WRITING PLAN

Thesis title: Leveraging Large Language Models for Personality Trait Detection with ICL and Retrieval-Augmented Techniques

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Week	Period	Detailed Tasks & Deliverables
Week 1 · May 18 – 24 | Target: **Chapter 3: Methodology** fully drafted (8–15 pages) + 3 pipeline figures		
Week 1	**Mon – Tue** May 18 – 19	• Set up writing project from the template; build outline with all 6 chapters, and refs.bib (seed with PADO, P2P, BERT, SBERT, RAG, CoT, nomic-embed-text, NEO-PI-R). • Write **Chapter 3: Methodology — 3.1 Overview**: narrate Phase 1 (index build), Phase 2 (inference) pipeline (~1 page). Produce Figure 3.1 — full pipeline architecture diagram. • Write **3.2 30-Facet Psychological Profiler** (~2 pages):. Insert Table 3.1 — 30 NEO-PI-R facets.
	Wed – Thu May 20 – 21	• Write **3.3 Dual & Sliced-Dual Embedding**: alpha-weighted fusion equation; raw post vs. profile embedding trade-offs (~1.5 pages). Source: rag/embedder.py. • Write **3.4 Trait-Aligned Facet Vector & Hybrid Retrieval**: score = beta*dense + gamma*facet equation; explain why gamma > beta; trait-masked 6-dimensional slice (~1.5 pages). Source: rag/facet_vector.py, rag/retriever.py. • Write **3.5 SBERT Fine-Tuning**: multiple-negatives ranking loss on same-label triplets, anti-leakage split, evaluation on validation triplets.
	Fri – Sat May 22 – 23	• Write **3.6 Prompt Designs**: show all prompt templates verbatim in monospace boxes — Zero-shot, One-shot, CoT, Reasoned-RAG (~4 pages). Source: baselines/prompt.py, ptd_model/prompts.py. • Write **3.7 Label-Balanced Retrieval Policy**: interleave high/low buckets at retrieval time. Write **Chapter 3: Methodology** closing paragraph linking to **Chapter 4: Experiments & Evaluation**. • Produce Figure 3.2 — facet vector encoding example (essay -> 30-d ordinal vector -> 6-d trait-masked). Generate Figure 3.3 — dataset preparation flow (Essays + myPersonality -> train/val/test splits).
	Sun May 24	• Review **Chapter 3: Methodology** — check page count (8–15 p), all equations/figures/tables referenced in prose, no bullet points in body text.
Week 2 · May 25 – 31 | Target: **Chapter 4: Experiments & Evaluation** fully drafted (~12 pages); all result tables and plots locked		
Week 2	**Mon – Tue** May 25 – 26	• Write **4.1 Experimental Setup**: hardware, gpt-4o-mini and LLaMA-3-8B endpoints, OpenAI API budget, temperature=0, seed.

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		• Write **4.2 Datasets**: Essays corpus (2 469 docs, train=1 975 / val=247 / test=247 / test50=50) and myPersonality (245 users, train=194 / test=50); label balance per trait. • Write **4.3 Evaluation Metrics** with equations: accuracy, macro-F1, weighted-F1, per-class precision/recall, retrieval hit-rate and mean-match-rate. Build Table 4.1 — dataset statistics by corpus.
	Wed – Thu May 27 – 28	• Write **4.4 Retrieval Ablation**: compare rawpost vs. profile vs. sliced-dual vs. full-dual vs. hybrid-facet (gamma sweep). Pull numbers from result/retrieve_accuracy/*/accuracy_summary.csv. • Build Table 4.2 — 5 retrieval strategies x 5 traits x {k=3, k=5} -> mean-match-rate and hit-rate. Build Figure 4.1 — bar chart: mean-match-rate per strategy per trait at k=3. • Write **4.5 Prompting Ablation**: Zero-shot / One-shot baseline -> RAG -> RAG-Def -> Reasoned-RAG progression. Build Table 4.3 — per-trait macro-F1 across 5 prompt modes. Build Figure 4.2 — grouped bar chart of macro-F1 per trait per prompt mode.
	Fri – Sat May 29 – 30	• Write **4.6 Bias Analysis** (~2 pages): compute predicted high/low ratio per trait from latest predictions.csv; produce Figure 4.3 — predicted-ratio vs. ground-truth ratio per trait; discuss anti-bias clause effect of Reasoned-RAG. • Write **4.7 Error Analysis**: pick 4–6 misclassified essays from reasoning_log.jsonl; show evidence/counter-evidence/verdict/label and explain failure modes (~2 pages). Source: result/gpt-4o-mini/reasoned_rag_def_oneshot/20260514-220745/reasoning_log.jsonl. • Write **Chapter 4: Experiments & Evaluation** closing paragraph. Lock all result CSVs — no re-runs after today.
	Sun May 31	• Review **Chapter 4: Experiments & Evaluation** — verify every number traces to a CSV, page count ≤ 14 p, all figures/tables referenced.
Week 3 · Jun 1 – 7 | Target: **Chapter 2: Background & Related Work** + **Chapter 1: Introduction** + **Chapter 5: Solution & Final Contribution** + **Chapter 6: Conclusion** drafted		
Week 3	**Mon – Tue** Jun 1 – 2	• Write **Chapter 2: Background & Related Work** — **2.1 Scope of Research**: binary high/low classification for 5 OCEAN traits, text-only input. • Write **2.2 Related Work** — 6-paper comparison table (Table 2.1: model, dataset, approach, F1, limitation): PADO (personality-aware dialogue optimization), P2P (Personality to Prediction), BERT (Devlin et al.), RoBERTa (Liu et al.), SVM/LSTM baselines (Mairesse et al.), plus 1 RAG-for-classification baseline. • Write **2.3 Big Five & NEO-PI-R 30 Facets**: personality model (Costa & McCrae), trait-facet hierarchy, linguistic correlates (LIWC by Pennebaker & King).

Week	Period	Detailed Tasks & Deliverables
	Wed – Thu Jun 3 – 4	• Write **2.4 LLM Prompting Paradigms**: zero-shot, few-shot ICL, CoT (Wei et al.), RAG (Lewis et al.), and their trade-offs for personality classification. • Write **2.5 Dense + Hybrid Retrieval**: FAISS dense retrieval, SBERT (Reimers & Gurevych) and nomic-embed-text-v1.5 encoders, DPR (Karpukhin et al.), facet-aware re-ranking. • Write **Chapter 1: Introduction — 1.1 Problem Statement**, **1.2 Background & Problems of Research**, **1.3 Research Objectives & Conceptual Framework** (~2.5 pages). Draft Figure 1.1 — high-level conceptual diagram.
	Fri – Sat Jun 5 – 6	• Write **1.4 Contributions** (4 numbered bullets, ≤ 25 words each) and **1.5 Organization of Thesis** (paragraph form, no bullets). Self-review **Chapter 1: Introduction** against template '3–6 pages' rule. • Write **Chapter 5: Solution & Final Contribution — 5.1 Integrated Pipeline** (best config: Reasoned-RAG-Def-Oneshot, hybrid retrieval k=3, per-trait beta/gamma), **5.2 Comparison with Literature** (Table 5.1 vs. PADO, P2P, BERT), **5.3 Limitations** (bias drift, narrow corpora, single-model dependency). • Write **Chapter 6: Conclusion — 6.1 Summary** (restate what was solved; do NOT copy intro) and **6.2 Future Work** (bigger LLMs, regression instead of binary, multilingual extension, profiler continued pre-training).
	Sun Jun 7	• Review **Chapter 1: Introduction**, **Chapter 2: Background & Related Work**, **Chapter 5: Solution & Final Contribution**, and **Chapter 6: Conclusion** — check page counts, cross-references, and citation tags.
Week 4 · Jun 8 – 14 | Target: Abstract, front matter, references, appendices, final polish — SUBMISSION		
Week 4	**Mon – Tue** Jun 8 – 9	• Draft Abstract (200–350 words) — 4-part structure: (i) problem & gap, (ii) chosen approach & why, (iii) solution overview, (iv) main contributions & key result numbers. • Write Acknowledgment (100–150 words, following template). • Build List of Abbreviations (OCEAN, LLM, RAG, FAISS, SBERT, NEO-PI-R, ICL, CoT, F1, PLM, DPR…), List of Figures, List of Tables, and Table of Contents (auto-generate from cross-references).
Week 4	**Wed – Thu** Jun 10 – 11	• Resolve every [CITE:xxx] tag — use Google Scholar to grab canonical BibTeX for each entry. Format all references per IEEE template style (no Wikipedia, no slides). Confirm >= 20 references, all cited in body; remove uncited entries. • Write **Appendix A: Reproducibility** — environment, package versions, random seed, gpt-4o-mini API config. • Write **Appendix B: Prompt Templates** — verbatim copies of SYS_PROMPT, SYS_PROMPT_REASONED, and all USR prompts. Source: ptd_model/prompts.py, baselines/prompt.py.

Week	Period	Detailed Tasks & Deliverables
		• Write **Appendix C: Classification Reports** — per-trait per-mode full reports.
	Fri – Sat Jun 12 – 13	• Whole-manuscript style pass — no bullet points in body text, every figure/table captioned and referenced in prose, every paragraph one main idea. • Replace all absolute/non-academic language with neutral phrasing. Run extract-text on final .docx to spot encoding/font issues. • Self-check against template Appendix A.1 General Regulations checklist. Cross-reference pass — every cross-link resolves; no 'Table ??' or 'Figure ??'. Page-number, header, footer pass. • Build final PDF; spot-check every page in a reader. Send draft PDF to supervisor for one final read.
	Sun Jun 14	• Review and submit full thesis

Milestone Summary by End of Each Week

End of Week 1 — May 24:

- **Chapter 3: Methodology** final draft (8–15 pages): **3.1 Overview** through **3.7 Label-Balanced Retrieval Policy** + Figures 3.1, 3.2, 3.3 + Table 3.1

End of Week 2 — May 31:

- **Chapter 4: Experiments & Evaluation** final draft (~12 pages): **4.1–4.7** + Tables 4.1, 4.2, 4.3 + Figures 4.1, 4.2, 4.3 — all result CSVs LOCKED

End of Week 3 — Jun 7:

- **Chapter 2: Background & Related Work** final draft (≤ 10 pages): **2.1–2.5** + Table 2.1 (Related Work Comparison)
- **Chapter 1: Introduction** final draft (3–6 pages): **1.1–1.5** + Figure 1.1
- **Chapter 5: Solution & Final Contribution** final draft (~4 pages) + Table 5.1
- **Chapter 6: Conclusion** final draft (1–2 pages)

End of Week 4 — Jun 14 (SUBMISSION):

- Abstract (200–350 words)
- References (>= 20 entries, IEEE style, all cited in body)
- **Appendix A: Reproducibility** + **Appendix B: Prompt Templates** + **Appendix C: Classification Reports**
- Submitted thesis PDF